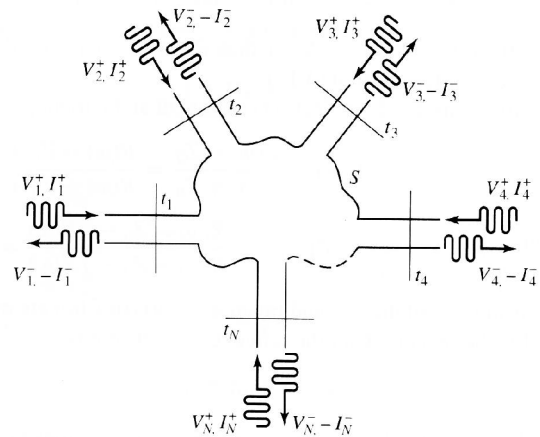


Wave Ports and Lumped Terminals

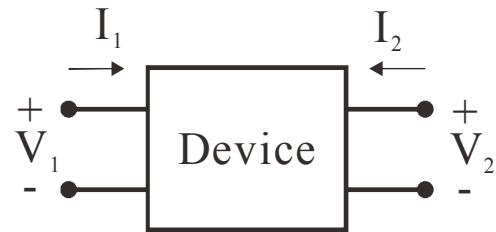
Wave Port

1. The device must be connected by a section of transmission line or waveguide supporting traveling waves.
2. The length of the line must be long enough such that only one propagation mode exists on the reference plane of the port.
3. Generalized S-parameters are calculated directly. Other parameters such as impedance matrices are converted from S-parameters mathematically.
4. Characteristic impedances of the transmission lines or waveguides might not be defined.



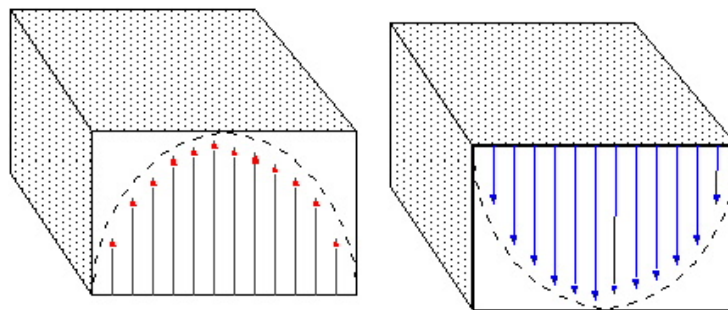
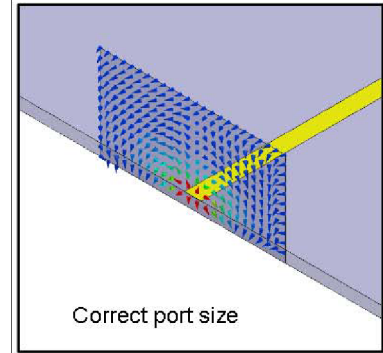
Lumped Terminal

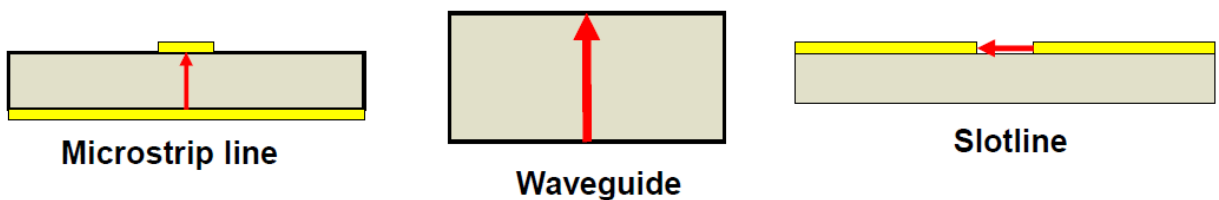
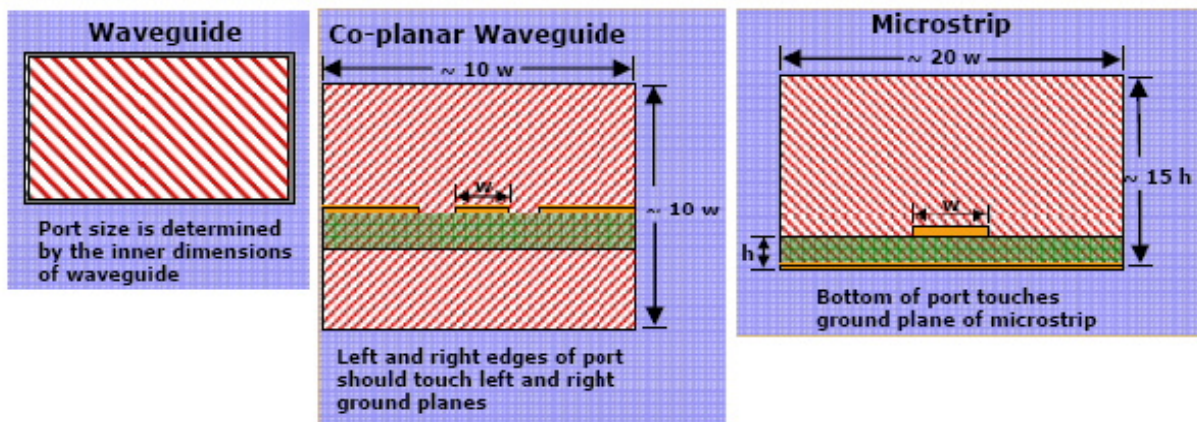
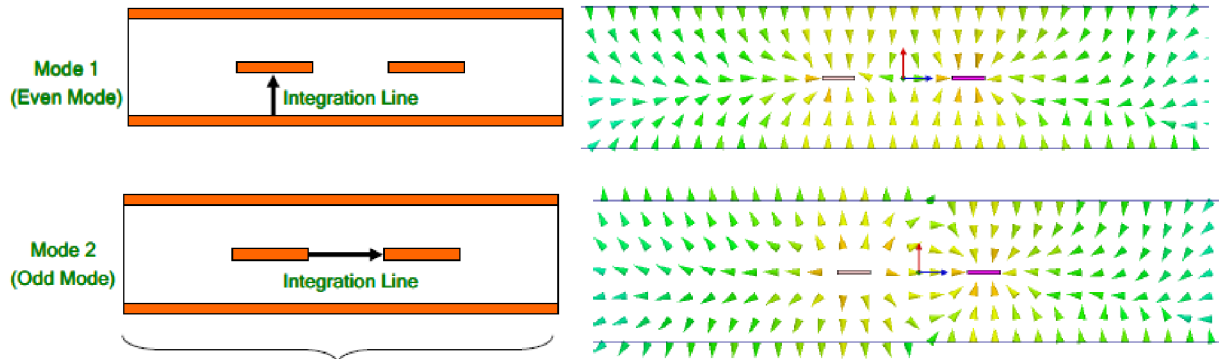
1. Both terminals must be connected to metal.
2. The structure is excited by a fix current I . Then, the electric field is solved and integrated across the terminal to find V . The impedance is computed by V/I . Depending on the formulation, it is also possible to use a fixed V to excite the structure, then compute the resulting current.
3. Impedance matrices are calculated directly. Other parameters such as S-parameters are converted from impedance matrices mathematically with user selected port impedance.
4. Only accurate when the distance between the two terminals is small compare to wavelength.



Wave Ports in HFSS

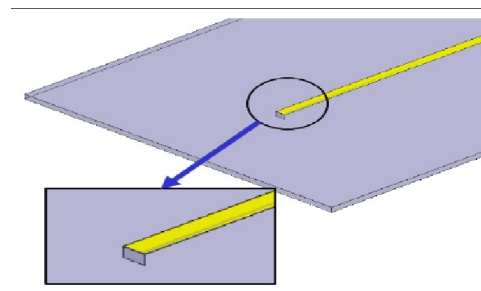
1. Only on external boundary.
2. A two-dimensional eigenvalue problem is solved first to find the waveguide modes of this port. The modal complex propagation constants and characteristic impedances are computed.
3. The mode patterns are used as the excitation.
4. Generalized S-parameters are computed by matching the fields on the port to the mode pattern. If higher-order modes exist on the boundary, this process may be contaminated, leading to wrong S-parameters. Therefore, it is necessary to keep a distance to the device under test.
5. For transmission line problems, such as microstrip lines or CPW, in theory, the size of the wave port should be as large as the boundary it touches. In reality, it can be smaller than the boundary to accommodate more than one port on one side, to solve antenna problems, or to avoid waveguide modes.
6. Use integration line to align the right mode patterns to make S-parameter computation consistent.
7. Characteristic impedances are computed according to integration line.
8. De-embed is possible due to the computed complex propagation constants ($\gamma = \alpha + j\beta$, $\Gamma' = \Gamma e^{-2\gamma L}$).





Lumped Port in HFSS

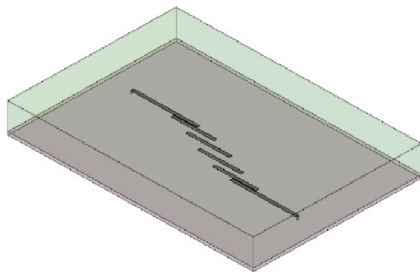
1. Can be internal.
2. No de-embedding.
3. An integration line also must be specified to indicate the path of electric field integration.
4. Converted to S-parameters by a port impedance supplied by



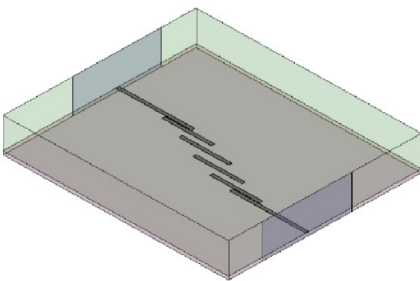
the user.

5. The results of lumped ports and wave ports are **never** the same.

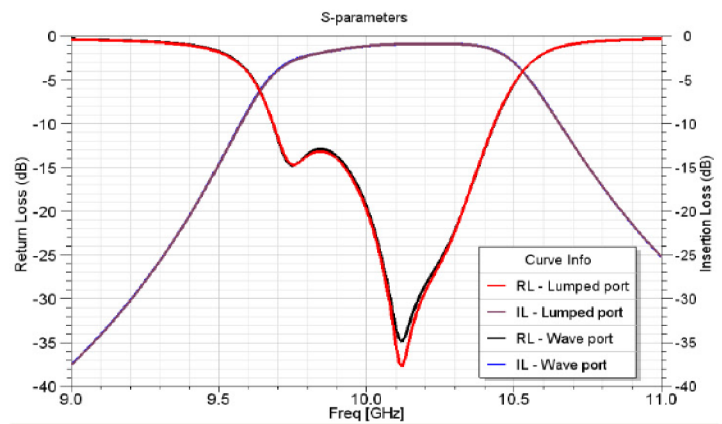
	Wave port	Lumped port
Location	External boundary	Internal
Higher order modes	Yes	No
De-embedding	Yes	No
Re-normalizing	Yes	Yes
Set-up complexity	Moderate	Low



Lumped Ports

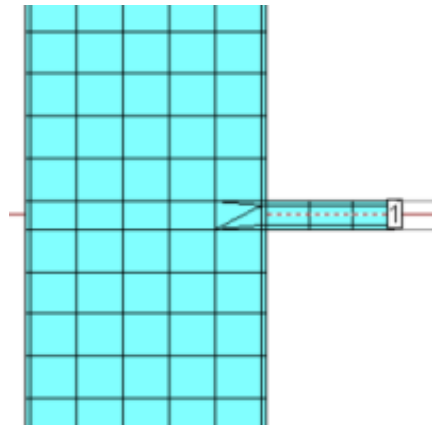


Wave Ports



Wave Ports in IE3D

1. Defined at the edge of transmission line. The line is automatically extended. The extension section **cannot** touch other traces.
2. A gap source is placed at the end of the extended section to excite a propagation wave along the transmission line.
3. Extension length can be adjusted. Insufficient extension length leads to wrong results.
4. S-parameters are computed from the VSWR on the line. Guided wavelengths can be found also from the maximum or minimum of VSWR.
5. De-embedded is possible.
6. Possible to excite higher-order propagation modes.
7. The wave port in IE3D is called Extension for Waves.
8. 50-Ohms for Waves is the same as Extension for Waves except S-parameters are converted mathematically for 50 Ω port impedance.



Lumped Ports in IE3D

1. Advanced Extension (lumped): The excitation is the same as wave port, but impedance is computed by V/I where I is the total current on the transmission line and V is the voltage from the line to ground computed by a suitable integrating scheme. S-parameters are converted from impedance matrix by assume 50 Ω port impedance.
2. Extension for MMIC (lumped): Same as Advanced Extension except using a different integration scheme. Outdated.
3. Localized for MMIC (conventional lumped)
4. Vertical Localized (conventional lumped).
5. Horizontal Localized (conventional lumped).

Coupled Line Structures

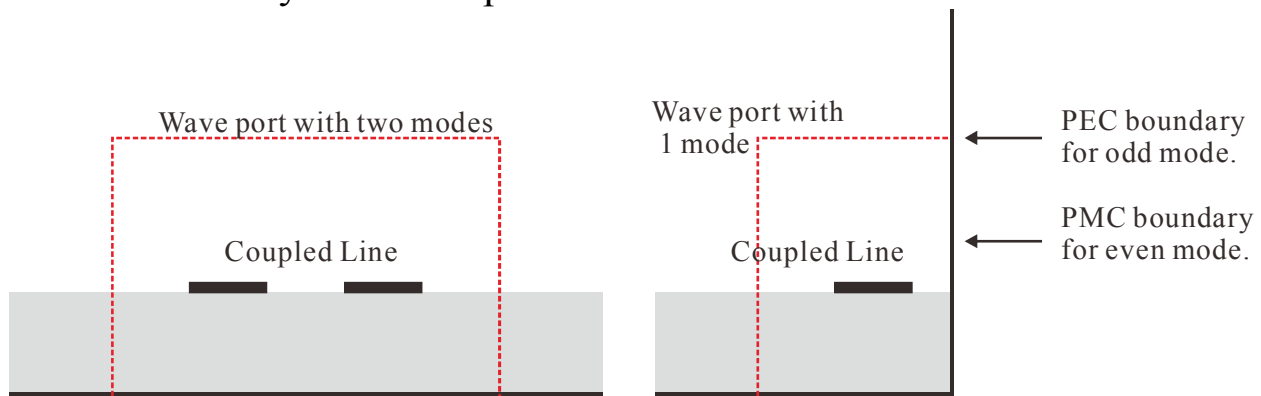
Difficulty:

1. Traces are too close to cause port coupling if ports are placed at the end of the traces.
2. Bends can be added to separate the ports at the cost of extra effects.



To simulate the coupled line characteristics without the effect of bends:

1. HFSS: Recombine the even-odd mode 2-port S-parameters to form 4-port S-parameters. Two way to perform even-odd mode analysis:
 - a. Select 2 modes (even-odd) per wave port.
 - b. Use symmetry by setting PEC or PMC boundary condition at the symmetrical plane.



2. IE3D: Use even-odd mode excitation to combined ports.

